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karthik@Karthikeya:~$ nano process_states.c
karthik@Karthikeya:~$ gcc process_states.c -o process_states
./process_states
=== Process State Demonstration ===
[Initial Parent] PID: 1117, PPID: 308 | State: R (running)

-- Child Process Started --
[Child Running] PID: 1118, PPID: 1117 | State: R (running)
Child entering sleeping state (sleep 3s)...
[Parent Observing Child] PID: 1118, PPID: 308 | State: S (sleeping)
[Child Resumed] PID: 1118, PPID: 1117 | State: R (running)
--- Child Exiting ---

--- Child Terminated ---
[Parent After wait()] PID: 1117, PPID: 308 | State: R (running)
karthik@Karthikeya:~$ nano task2.c
karthik@Karthikeya:~$ gcc task2.c -o task2
./task2 &
[1] 1141
karthik@Karthikeya:~$ Process started
PID : 1141
PPID: 308
Process State: Running
Process State: Sleeping/Waiting
Process State: Running again
Process State: Terminating
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[1] ~ Done
karthik@Karthikeya:~$ ps -o pid,ppid,stat,cmd -C task2
  PID   PPID  STAT  CMD
karthik@Karthikeya:~$ kill -STOP $(pgrep task2)
kill: usage: kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l [sigspec]
karthik@Karthikeya:~$ ps -o pid,ppid,stat,cmd -C task2
  PID   PPID  STAT  CMD
karthik@Karthikeya:~$ kill -CONT $(pgrep task2)
kill: usage: kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l [sigspec]
karthik@Karthikeya:~$ ps -o pid,ppid,stat,cmd -C task2
  PID   PPID  STAT  CMD
```